Assignment Day 3 (28/02/2026)

Assignment 1

Problem Statement: Array Operations using Class in C++

Design a C++ program using Object-Oriented Programming concepts to perform basic operations on an integer array.

Create a class named ArrayOperations that contains data members and member functions to perform the following tasks:

1. Accept n integer elements from the user.
2. Display the array elements.
3. Find and display the maximum and minimum elements.
4. Calculate and display the sum and average of elements.
5. Search for a given element in the array and display its position (if found).

Additional Requirements:

- The array size should be taken from the user.
- Use a menu-driven approach.
- Do not use STL (like vector).

Objective:

To understand array handling, searching logic, and class-based programming in C++.

Solution:

Project Location:

D:\Vikas\Day3\ArrayOperation

file locations

Header File

D:\Vikas\Day3\ArrayOperation\ArrayOps.h

```
#pragma once
#include<iostream>
using namespace std;

class ArrayOps {
private:
	int noe;
	int* arr;

public:
	ArrayOps(); //Default Constructor
	void accept(); //Accept value from user
	void display(); //Display Elements of Array
	void findMax(); //Find Maximum Value
	void findMin();     // FInd Minimum Value
	void sumOfNumbers();      // Sum of All element
	void avgNum();      // Average of all num
	void searchElement();     // Serch element function
};
```

Resource File

D:\Vikas\Day3\ArrayOperation\ArrayOps_src.cpp

```
#include"ArrayOps.h"


ArrayOps::ArrayOps() {

	// Allocate array
	cout << "\nEnter the number of element : ";
	cin >> noe;

	arr = new int[noe];

	// Initialize array
	for (int i = 0; i < noe; i++)
	{
```

```
            arr[i] = 0;
        }
}

// Accept Input from user
void ArrayOps::accept() {
        for (int i = 0; i <noe; i++)
        {
            cout << "\nEnter the Number : ";
            cin >> arr[i];
        }
}

// Display array output
void ArrayOps::display(){
        for (int i = 0; i < noe; i++)
        {
            cout << "\n The Element is : "<<arr[i];

        }
}

// Display Maximum number
void ArrayOps::findMax(){
        int max = arr[0];

        for (int i = 0; i < noe; i++) {
            if (arr[i] > max) {
                max = arr[i];
            }
        }
        cout << "\nMaximum num is : " << max;

}

//Display Minimum number
void ArrayOps:: findMin() {
        int min = arr[0];

        for (int i = 0; i < noe; i++) {
            if (arr[i] < min) {
                min = arr[i];
            }
        }
        cout << "\nMinimum num is : " << min;
}
\
//Display Sum of all number
void ArrayOps::sumOfNumbers() {
        int sum = 0;
        for (int i= 0; i < noe; i++) {
            sum += arr[i];
        }
        cout << "\nSum of num is : " << sum;
}

//Display Average number
void ArrayOps::avgNum() {
```

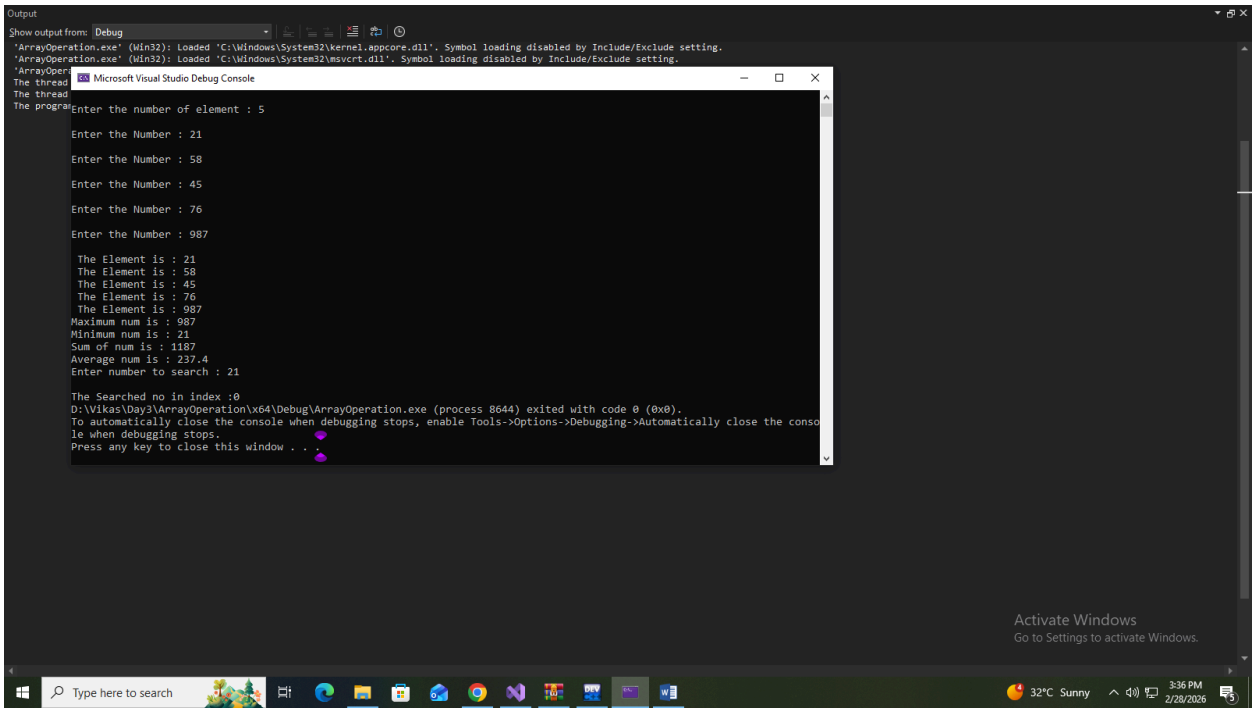
```
	float sum = 0, avg;
	for (int i = 0; i < noe; i++) {
		sum += arr[i];
	}
	avg = sum / noe;
	cout << "\nAverage num is : " << avg;
}

//Display number index in array
void ArrayOps::searchElement() {
	int n1;
	cout << "\nEnter number to search : ";
	cin >> n1;
	for (int i = 0; i < noe; i++) {
		if (arr[i] == n1) {
			cout << "\nThe Searched no in index : " << i;
		}
	}
}
```

Client/Main file

D:\Vikas\Day3\ArrayOperation\Client.cpp

```
#include"ArrayOps.h"

int main() {

	ArrayOps obj1;

	// Accept Value
	obj1.accept();

	//Display Value
	obj1.display();

	//Diplay max value
	obj1.findMax();

	// display minimum value
	obj1.findMin();

	// display total number
	obj1.sumOfNumbers();

	// display average number
	obj1.avgNum();

	//display search element index num
	obj1.searchElement();
}
```

Output


Output
Show output from: Debug
'ArrayOperation.exe' (Win32): Loaded 'C:\Windows\System32\kernel.appcore.dll'. Symbol loading disabled by Include/Exclude setting.
'ArrayOperation.exe' (Win32): Loaded 'C:\Windows\System32\msvcrt.dll'. Symbol loading disabled by Include/Exclude setting.
Microsoft Visual Studio Debug Console
Enter the number of element : 5
Enter the Number : 21
Enter the Number : 58
Enter the Number : 45
Enter the Number : 76
Enter the Number : 987
The Element is : 21
The Element is : 58
The Element is : 45
The Element is : 76
The Element is : 987
Maximum num is : 987
Minimum num is : 21
Sum of num is : 1187
Average num is : 237.4
Enter number to search : 21
The Searched no in index :0
D:\Vikas\Day3\ArrayOperation\x64\Debug\ArrayOperation.exe (process 8644) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
Activate Windows
Go to Settings to activate Windows.
Type here to search
32°C Sunny
3:36 PM
2/28/2026

Assignment 2

Problem Statement: String Operations using Class in C++

Develop a C++ program using a class to perform various string manipulation operations without using built-in string library functions (except input/output).

Create a class named StringOperations that performs the following:

1. Accept a string from the user.
2. Display the entered string.
3. Find the length of the string manually (without using strlen()).
4. Reverse the string.
5. Check whether the string is a palindrome.
6. Convert all lowercase characters to uppercase.

7. Compare two strings without using strcmp().

Additional Requirements:

- Use character arrays (char[]).
- Implement logic manually using loops and conditions.
- Use a menu-driven program.

Solution